SYNTH-04: Private Synthetic Locations

> **Series:** SYNTH — Synthetic Monitoring | **Notebook:** 4 of 6 | **Created:** December 2025 | **Last Updated:** 10/19/2026

Monitoring Internal Applications from Your Infrastructure

This notebook covers deploying and managing private synthetic locations (ActiveGates) for monitoring internal applications, APIs, and services not accessible from the public internet.

Table of Contents

1. Why Private Locations?
2. Architecture
3. Synthetic-Enabled ActiveGate
4. Deployment Options
5. Configuration
6. Monitoring Private Location Health
7. Troubleshooting

Prerequisites

- ✅ Access to a Dynatrace environment with Synthetic Monitoring

- ✅ Completed SYNTH-01 through SYNTH-03
- ✅ Infrastructure access to deploy ActiveGate (for setup)

1. Why Private Locations?

Public vs Private Locations

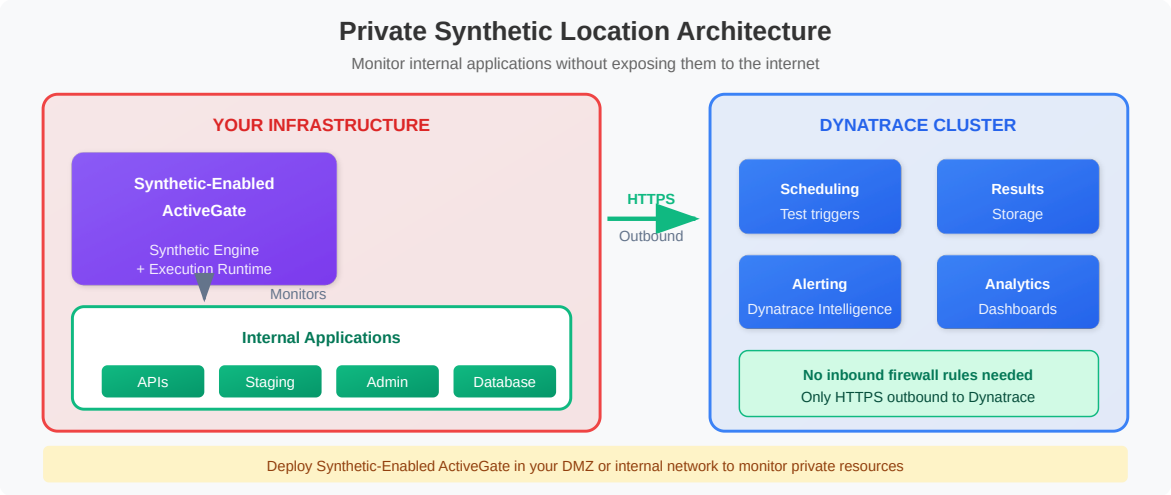
Aspect	Public Locations	Private Locations
Hosting	Dynatrace cloud	Your infrastructure
Access	Public internet only	Internal networks
Maintenance	Managed by Dynatrace	Managed by you
Security	External perspective	Behind firewall
Latency	Varies by region	Local network

Use Cases for Private Locations

Scenario	Description
Internal APIs	Services not exposed to internet
VPN-only apps	Applications requiring VPN access
Pre-production	Staging/dev environments
Security compliance	Data must not leave network
Low latency testing	Local performance baselines
Isolated networks	Air-gapped or restricted networks

2. Architecture

Private synthetic locations use ActiveGates deployed within your infrastructure to execute monitors against internal applications:



Key Points

- **Outbound only**: ActiveGate initiates all connections
- **No inbound ports**: No firewall changes for external access
- **Local execution**: Tests run inside your network
- **Results upload**: Only metrics/results sent to Dynatrace

3. Synthetic-Enabled ActiveGate

ActiveGate Capabilities

ActiveGate can serve multiple purposes:

Capability	Description
Synthetic	Private synthetic location
Metrics	Metric ingestion endpoint
Logs	Log ingestion endpoint
API	Cluster API proxy
Extensions	Extension execution

Synthetic Engine Requirements

Resource	Minimum (XS node with browser support)	Guidance
CPU	2 vCPU	4 vCPU for an S node — pick the node size from the sizing guide's executions-per-hour limits
RAM	4 GB	At least 8 GB
Disk	20 GB free	At least 25 GB free
Network	HTTPS outbound	Low latency to targets
Operating system	A Linux or Windows Server release currently supported for a synthetic-enabled ActiveGate	Track the ActiveGate OS support matrix — supported releases are versioned with the **ActiveGate**, not with the tenant

The 8 GB / 25 GB guidance is the requirements page's own: *"we recommend having at least 8 GB of RAM and 25 GB of free disk space."* Size is also sticky — *"To change the size of a Synthetic-enabled ActiveGate, for example, after upgrading from size S to meet size M requirements, you must uninstall and reinstall it."*

Windows Server 2025 is supported for private synthetic locations. The OS option first appeared for synthetic-capable ActiveGates with **ActiveGate 1.341**; **SaaS 1.344** (release notes published 07/27/2026) states it in the private-location context directly: *"You can now install on Windows Server 2025 hosts for private synthetic locations."* The documentation statement is firm — published docs are live regardless of which sprint your tenant is on — but the capability ships in the **ActiveGate installer**, so confirm the ActiveGate build you are about to deploy carries it before committing a Windows Server 2025 rollout plan.

ActiveGate **1.343** (published 07/15/2026, rollout from 07/28/2026) additionally adds OS support for **Oracle Linux 9.8 and 10.2** and **Rocky Linux 9.8 and 10.2**, and flags a support-discontinuation wave. The ActiveGate 1.347 notes give the dates: Red Hat Enterprise Linux 9.4 and Ubuntu 16.04 from 01 November 2026, Amazon Linux 2 from 01 January 2027, and Oracle Linux 9.8 / 10.2 and Rocky Linux 9.8 / 10.2 — the releases 1.343 added — *"The following operating systems will no longer be supported starting 01 June 2027"*. Audit your private-location fleet against those notes before the first date.

Browser Monitor Requirements

For browser monitors, additional requirements:
- A Chromium-family browser, provided through the ActiveGate rather than installed by you — on **Windows** the installer package includes Chrome for Testing; on **Linux** the installer downloads the browser and its dependencies. Its version tracks your **ActiveGate fleet** (and, on Linux, the location's browser auto-update switch — see below), **not your tenant version**
- Write access to `/tmp` — the browser's dependencies, including `xvfb`, are installed with it and use `/tmp` (no separate display server to install)
- Additional RAM for browser instances

Browser baseline by ActiveGate version:

ActiveGate	RHEL 9.7, Rocky Linux 9.8	Ubuntu 20.04 / 22.04 / 24.04, Amazon Linux 2023, Oracle Linux 9.7	Windows (bundled in the installer)
1.343 (published 07/15/2026, rollout from 07/28/2026)	Chromium 150	Chrome for Testing 150	Chrome for Testing 150
1.345 (rollout from 08/25/2026)	Chromium 151	Chrome for Testing 151	Chrome for Testing 151
1.347 (published 10/01/2026, rollout from 09/30/2026)	Chromium 152	Chrome for Testing 152	Chrome for Testing 152

Fleets move to each row as their ActiveGates update — check the ActiveGate version (query below) before assuming Chromium 152.

How the browser is updated. The browser moves with ActiveGate and Synthetic engine updates, not on a schedule of its own: *"The browser autoupdate takes place during manual as well as automatic ActiveGate and Synthetic engine updates."* On **Linux** this is controlled per private location by the **Enable Chrome(-ium) auto-update** switch, which is on by default (*"the browser autoupdate is turned on by default for locations with Linux-based ActiveGates"*). Turn it off only to hold a specific browser version or for offline environments — and then update the browser manually on every ActiveGate in the location. On **Windows** there is no switch: *"on Windows-based ActiveGates, the browser is always updated during Synthetic engine updates."* So the lever for holding a browser version is that per-location switch, **not** ActiveGate auto-update. Two support rules bound

how far a location can drift: Dynatrace *"supports browser versions that are no more than two versions behind the latest Dynatrace-supported version for a specific ActiveGate release"*, and *"We strongly recommend updating all ActiveGates per location to the same version."*

Because the browser can change whenever an ActiveGate or engine update lands, re-check browser monitors after an update before treating new failures as application regressions.

> **Sources:** ActiveGate 1.343 (DT docs) — *"Chrome for Testing 150 is bundled with the Windows ActiveGate installer."*; ActiveGate 1.345 (DT docs) — *"Chrome for Testing 151 is bundled with the Windows ActiveGate installer."*; ActiveGate 1.347 (DT docs) — *"Chromium 152 is the latest supported version for Synthetic-enabled ActiveGate"* and *"Chrome for Testing 152 is bundled with the Windows ActiveGate installer."*; Requirements for private Synthetic locations (DT docs) — *"Its dependencies, including xvfb, utilize /tmp"*; Manage private Synthetic locations (DT docs) — the auto-update, Windows, two-versions-behind and same-version quotes above.

Two things follow from the browser being tied to the ActiveGate. First, an ActiveGate that has not been updated — or a Linux location with browser auto-update turned off — is still executing clickpaths in the older browser however current the tenant is — so when the *same* clickpath behaves differently at two private locations, compare **ActiveGate versions before** suspecting the application:

```
smartscapeNodes "ACTIVEGATE"
| fields name, dt.active_gate.version, dt.active_gate.group.name, os.type,
os.version
| sort dt.active_gate.version asc, name asc
```

Second, the **1.331 floor stated below is a minimum, not a target.** It is the version that lets users whose access is scoped by security context see private-location results; the ActiveGate build (and, on Linux, the location's browser auto-update switch) determines which browser your clickpaths actually run in. Both matter, for different reasons. (For the ActiveGate update-management model on SaaS — auto-update windows, version pinning, staged fleet upgrades — see FAQ-05; the browser itself is governed by the per-location switch above.)

> **Security context and ActiveGate 1.331 (SaaS 1.343, July 2026 — staged tenant rollout from mid-July).** SaaS 1.343 migrates the management zones of each

> synthetic **monitor** to security-context values: *"Dynatrace performs a one-time migration of the management zone each synthetic monitor belongs to, mapping them to security context values with the same name on that monitor."* It runs once — *"Updating or creating management zones and monitors won't be synchronized."* On private locations, ActiveGate **1.331+** is what makes those results scopable: *"Earlier versions do not enrich the metrics and events produced by Synthetic monitor executions with the monitor's security context value. Without this enrichment, IAM policies scoped to security contexts cannot grant access to execution data."* An older ActiveGate keeps executing monitors; users whose access is scoped by security context just cannot see its results. Upgrade any synthetic AG below 1.331 before you move users to security-context-scoped policies, and review monitor access once the migration reaches your tenant — the security-context model replaces MZ-based scoping (the MZ2POL series covers the broader management-zone migration). SaaS 1.343 also adds IAM-based access control for Synthetic Monitoring and API support for assigning security context to synthetic monitors.
>
> **Sources:** SaaS 1.343 (DT docs), Access control for Synthetic (DT docs).

4. Deployment Options

Option 1: Linux/Windows Installer

Download the installer from **Settings** → **Deployment status** → **ActiveGate**, run it with the synthetic flag, then check the service:

```
sudo /bin/sh Dynatrace-ActiveGate-Linux-x86-*.sh \
  --enable-synthetic

sudo systemctl status dynatracegateway
```

Option 2: Kubernetes / OpenShift (containerized locations)

Containerized private locations are documented for **Kubernetes and OpenShift** only; there is no documented Docker / Docker Compose deployment for a synthetic-enabled ActiveGate.

Containerized private locations are deployed from **templates the Synthetic app generates**, not from a DynaKube custom resource — there is no `synthetic-monitoring`

ActiveGate capability to add to a DynaKube. In **Synthetic → Private locations**, create a containerized location and select **Download synthetic.yaml** (*"This is the location template file."*), then deploy the metric adapter from its own downloaded template (*"This is the template file for the Synthetic metric adapter."*). Apply both with the `kubectl` commands the UI generates.

> ⚠️ **Latest Dynatrace — metric adapter (Sprint 1.339):** the variable belongs on the **Synthetic metric adapter** deployment, not on the location pod: *"Modify the adapter deployment. Set the environment variable METRIC_3RD_GEN_ENABLED to "true"."* The same migration requires you to *"Set the environment variable BASE_URL to match the Latest Dynatrace URL."* and to *"Modify synthetic location deployments—change the target metric in the Horizontal Pod Autoscaler definition."* Re-download the templates from **Synthetic → Private locations**, or patch existing manifests per those steps, and schedule the change through change management.
>
> **Sources:** Containerized private Synthetic locations on Kubernetes (DT docs).

```
// List all synthetic locations, public and private
// PREFERRED -- Smartscape. The SYNTHETIC_LOCATION node exposes location_type DIRECTLY,
// which makes the old "identify private locations by naming convention" workaround
// obsolete. `stage` is the lifecycle marker (GA / BETA / DEPRECATED / COMING_SOON) --
// worth scanning for locations Dynatrace is retiring under you.
smartscapeNodes "SYNTHETIC_LOCATION", from: now() - 7d  // default 2 h window can hide private locations
| fields name, location_type, stage, cloud.provider, geo.country.name, geo.city.name, id_classic
| sort location_type asc, name asc
| limit 50

// To list only private locations, add:
// | filter location_type == "PRIVATE"
//
// The literals observed on the validation tenant (10/02/2026) were "PUBLIC" and "PRIVATE";
// private locations carry no stage. Run the summarize query in the next cell against YOUR
// tenant before hard-coding a literal -- a wrong literal returns zero rows, not an error.

// FALLBACK (classic surface -- deprecated in DQL, supported for as long as Dynatrace
// Classic is supported, and genuinely lacks these fields.
// This is why the previous revision of this notebook said type/status/city/countryCode
// "are not available": on the classic entity that was true, and remains true.)
// fetch dt.entity.synthetic_location
// | fields id, entity.name
// | sort entity.name asc
// | limit 50
```

```
// Count synthetic locations by type and lifecycle stage
// Run THIS FIRST -- it tells you the exact location_type literals your tenant returns,
// which is what the previous cell's filter depends on.
smartscapeNodes "SYNTHETIC_LOCATION", from: now() - 7d
| summarize locations = count(), by:{location_type, stage}
| sort location_type asc, locations desc

// FALLBACK (classic surface, deprecated in DQL) -- can only produce a grand total, because the classic
// entity has no type field at all. That limitation is exactly what the old
// naming-convention workaround was compensating for.
// fetch dt.entity.synthetic_location
// | summarize {count = count()}
// | fields count
```

5. Configuration

Creating a Private Location

1. **Deploy ActiveGate** with synthetic capability
2. **Navigate to**: **Synthetic** → **Private locations** (or, in Settings Classic, **Settings** → **Web and mobile monitoring** → **Private Synthetic locations**)
3. **Create location**: Name, description, geographic info
4. **Assign ActiveGates**: Select which ActiveGates serve this location

Location Settings

Setting	Description
Name	Descriptive location name
Latitude/Longitude	Geographic coordinates
City/Region	Location metadata
ActiveGate nodes	Assigned ActiveGates — one ActiveGate cannot serve more than one location

Setting	Description
Enable Chrome(-ium) auto-update	Linux-based locations; on by default. Updates the browser during ActiveGate and Synthetic engine updates (see § 3). API property: `autoUpdateChromium`

High Availability

For production workloads:
- Deploy 2+ ActiveGates per location
- Keep every ActiveGate in a location on the same version
- Distribute across availability zones
- Executions are distributed across the location's ActiveGates

Declarative Creation via the API

The UI flow above is the interactive path. For repeatable, reviewable, source-controlled private-location definitions, use the Environment API v2 endpoint:

```
POST /api/v2/synthetic/locations
```

> **Platform alternative (early adopter).** `POST /platform/synthetic/v1/locations` creates a private location with a platform or OAuth token holding `synthetic:locations:write` — Dynatrace recommends migrating to it, and documents all of its endpoints as Early Adopter — they *"can change in non-compatible ways"*. The Environment API v2 endpoint above keeps working.
>
> **Sources:** POST locations — platform API (DT docs) — *"You use the locations endpoint to create a new private synthetic location."*; Access control — Synthetic (DT docs) — *"Dynatrace recommends migrating to the Synthetic platform permissions and the Synthetic Platform API"*; *"All Synthetic Platform API endpoints are currently in Early Adopter maturity."*; *"can change in non-compatible ways"*

API 1.344 (published 07/15/2026, **staged rollout from 07/29/2026**) adds three properties to the `PrivateSyntheticLocation` request schema of `POST /synthetic/locations` — `minActiveGateCount`, `maxActiveGateCount`, and `nodeSize`. All three are **containerized-location** properties: the POST reference describes each as a *"Containerized location property"* that is *"required for a Kubernetes location"*. They are the API counterparts of the fields the UI asks for when you create a Kubernetes or OpenShift location — the minimum

and maximum number of ActiveGates the horizontal pod autoscaler works between, and the ActiveGate node size. They do not size a standard Linux/Windows location, whose ActiveGate size is set by the host it is installed on (see § 3).

Property	Purpose
`minActiveGateCount`	Minimum number of ActiveGates — the autoscaler's lower bound. The docs recommend *"a minimum of two ActiveGates per location"*
`maxActiveGateCount`	Maximum number of ActiveGates — caps scale-out
`nodeSize`	ActiveGate node size: `XS`, `S` or `M` (*"The node size L is not supported in containerized locations."*). Choose deliberately — *"Once specified, ActiveGate size for a location can't be changed because persistent storage can't be resized."*

Because the rollout is staged, confirm the tenant you are automating accepts the three properties before relying on them; until it does, creating the containerized location in the UI remains the working path.

> **Authentication boundary:** the Environment API v2 endpoint above, and the Terraform provider path (AUTOM-04 § Authentication Boundary), use an **API token**. For new automation on Latest Dynatrace, *"Dynatrace recommends migrating to the Synthetic platform permissions and the Synthetic Platform API to manage Synthetic in Latest Dynatrace"*, which takes a bearer platform token. Check that the Synthetic Platform API covers the location operations you need before switching.
>
> **Sources:**
>
> - Dynatrace API release notes 1.344 (DT docs) — `POST /synthetic/locations`: changed `PrivateSyntheticLocation` schema, added `maxActiveGateCount`, `minActiveGateCount`, `nodeSize`
> - Synthetic locations API (DT docs)
> - POST a location (DT docs) — *"Containerized location property. The minimum number of ActiveGates deployed for the location (required for a Kubernetes location)."*
> - Containerized private Synthetic locations on Kubernetes (DT docs) — *"We recommend the S ActiveGate size and a minimum of two ActiveGates per location."*
> - Private locations in Synthetic (DT docs) — *"We recommend using at least two ActiveGates for a location. You can't use one ActiveGate for multiple locations."*
> - Access control (DT docs) — *"Dynatrace recommends migrating to the Synthetic platform permissions and the Synthetic Platform API to manage Synthetic in Latest Dynatrace."*

> - Manage private Synthetic locations (DT docs) — *"We strongly recommend updating all ActiveGates per location to the same version."*

```
// Monitor executions by location
// Smartscape fields: dt.entity.synthetic_location / dt.entity.synthetic_test
on synthetic data are
// deprecated ("will be removed in the future") in favor of dt.smartscape.* —
Synthetic events model, 09/21/2026.
// Private locations typically have custom names (not city names like "N.
Virginia")
fetch dt.synthetic.events, from: now() - 24h
| filter endsWith(event.type, "_monitor_execution")
| filter not(coalesce(execution.retry_on_error, false) and result.state ==
"FAIL")  // one result per scheduled run
| summarize {
    executions = count(),
    availability_pct = round(countIf(result.state == "SUCCESS") * 100.0 /
count(), decimals: 2)
  }, by: {monitor.name, dt.synthetic.monitor.id,
dt.smartscape.synthetic_location}
| fieldsAdd location = getNodeName(dt.smartscape.synthetic_location)
| sort executions desc
| limit 30
```

6. Monitoring Private Location Health

ActiveGate Health Indicators

Starting points from community practice — Dynatrace publishes no thresholds for these.

Metric	Description	Alert Threshold
Connectivity	Connection to cluster	Any disconnect
CPU Usage	Processing load	> 80% sustained
Memory	RAM utilization	> 80%
Disk	Storage usage	> 80%

```
// Synthetic location health status (1 = healthy)
// Smartscape fields: dt.entity.synthetic_location / dt.entity.synthetic_test
on synthetic data are
// deprecated ("will be removed in the future") in favor of dt.smartscape.* —
Synthetic events model, 09/21/2026.
// Surfaces location/ActiveGate availability independent of monitor results
timeseries health = avg(dt.synthetic.location.health_status),
    from: now() - 24h, interval: 1h, by: {dt.smartscape.synthetic_location}
| fieldsAdd location = getNodeName(dt.smartscape.synthetic_location)
```

```
// Location availability over time (last 7 days)
// Smartscape fields: dt.entity.synthetic_location / dt.entity.synthetic_test
on synthetic data are
// deprecated ("will be removed in the future") in favor of dt.smartscape.* —
Synthetic events model, 09/21/2026.
fetch dt.synthetic.events, from: now() - 7d
| filter endsWith(event.type, "_monitor_execution")
| filter not(coalesce(execution.retry_on_error, false) and result.state ==
"FAIL")  // one result per scheduled run
| makeTimeseries {
    success_count = countIf(result.state == "SUCCESS"),
    total_count = count()
  }, interval: 1h, by: {dt.smartscape.synthetic_location}
| fieldsAdd availability_pct = success_count[] * 100.0 / total_count[]
```

```
// Execution count by location
// Smartscape fields: dt.entity.synthetic_location / dt.entity.synthetic_test
on synthetic data are
// deprecated ("will be removed in the future") in favor of dt.smartscape.* —
Synthetic events model, 09/21/2026.
fetch dt.synthetic.events, from: now() - 24h
| filter endsWith(event.type, "_monitor_execution")
| filter not(coalesce(execution.retry_on_error, false) and result.state ==
"FAIL")  // one result per scheduled run
| summarize {
    total_executions = count(),
    failed_executions = countIf(result.state == "FAIL"),
    unique_monitors = countDistinct(dt.synthetic.monitor.id)
  }, by: {dt.smartscape.synthetic_location}
| fieldsAdd failure_rate = round((failed_executions * 100.0) /
total_executions, decimals: 2)
| fieldsAdd location = getNodeName(dt.smartscape.synthetic_location)
| sort total_executions desc
```

7. Troubleshooting

Common Issues

Issue	Symptoms	Resolution
No connectivity	Location offline	Check ActiveGate logs, network
SSL errors	Certificate failures	Install CA certs on ActiveGate
Timeouts	All tests fail	Check network routes, DNS
Resource exhaustion	Slow/failed tests	Scale ActiveGate resources
Browser issues	Clickpath failures	Check Chrome/display config

ActiveGate Logs

Linux default directories:

```
/var/log/dynatrace/gateway       ActiveGate logs
/var/log/dynatrace/synthetic     Private Synthetic logs
/var/tmp/dynatrace/synthetic     Private Synthetic temporary files (including
screenshots)
```

> **Sources:** ActiveGate directories (DT docs) — *"Private synthetic logs /var/log/dynatrace/synthetic"*.

Network Verification

```
# Test connectivity to target
curl -v https://internal-api.company.com/health

# DNS resolution
nslookup internal-api.company.com

# Test Dynatrace connectivity
curl -v https://&lt;tenant&gt;.live.dynatrace.com/api/v1/time
```

```
// Identify failing monitors by location
// Smartscape fields: dt.entity.synthetic_location / dt.entity.synthetic_test
on synthetic data are
// deprecated ("will be removed in the future") in favor of dt.smartscape.* —
Synthetic events model, 09/21/2026.
// To filter for one location, add: | filter dt.smartscape.synthetic_location
== toSmartscapeId("SYNTHETIC_LOCATION-...")
// (without toSmartscapeId() the comparison matches nothing — Grail only warns
SMARTSCAPEID_TO_STRING_COMPARISON)
fetch dt.synthetic.events, from: now() - 24h
| filter endsWith(event.type, "_monitor_execution")
| filter not(coalesce(execution.retry_on_error, false) and result.state ==
"FAIL")  // one result per scheduled run
| filter result.state == "FAIL"
| summarize {
    failure_count = count(),
    last_failure = max(timestamp)
  }, by: {monitor.name, dt.synthetic.monitor.id,
dt.smartscape.synthetic_location}
| fieldsAdd location = getNodeName(dt.smartscape.synthetic_location)
| sort failure_count desc
| limit 20
```

Summary

In this notebook, you learned:

✅ **Why private locations** - Internal apps, security, compliance
✅ **Architecture** - ActiveGate with synthetic engine
✅ **Deployment options** - Installer and Kubernetes/OpenShift templates (+ adapter `METRIC_3RD_GEN_ENABLED` for Latest tenants)
✅ **Configuration** - Creating and managing locations
✅ **Health monitoring** - `dt.synthetic.location.health_status` metric + execution results by location
✅ **Troubleshooting** - Common issues and resolution

Next Steps

Continue to **SYNTH-05: Network Monitoring** to learn about synthetic network availability monitoring.

References

- Private synthetic locations (DT docs)
- Synthetic architecture and communication (DT docs)
- ActiveGate (DT docs)
- Containerized private Synthetic locations on Kubernetes (DT docs)
- Requirements for private Synthetic locations (DT docs)
- Access control for Synthetic (DT docs)

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